

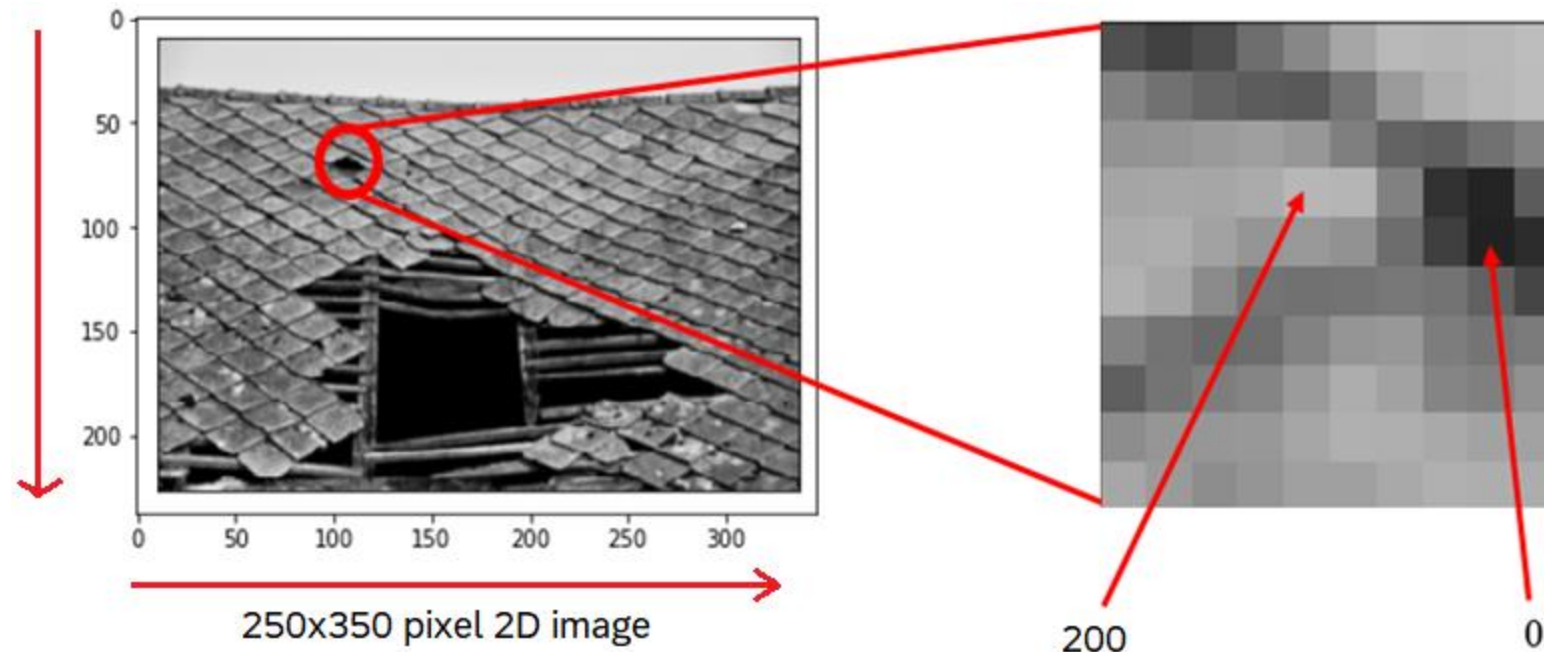
Welcome to Week 12

Computer Vision

Prof. Swapnil Kangralkar

Artificial Neural Networks

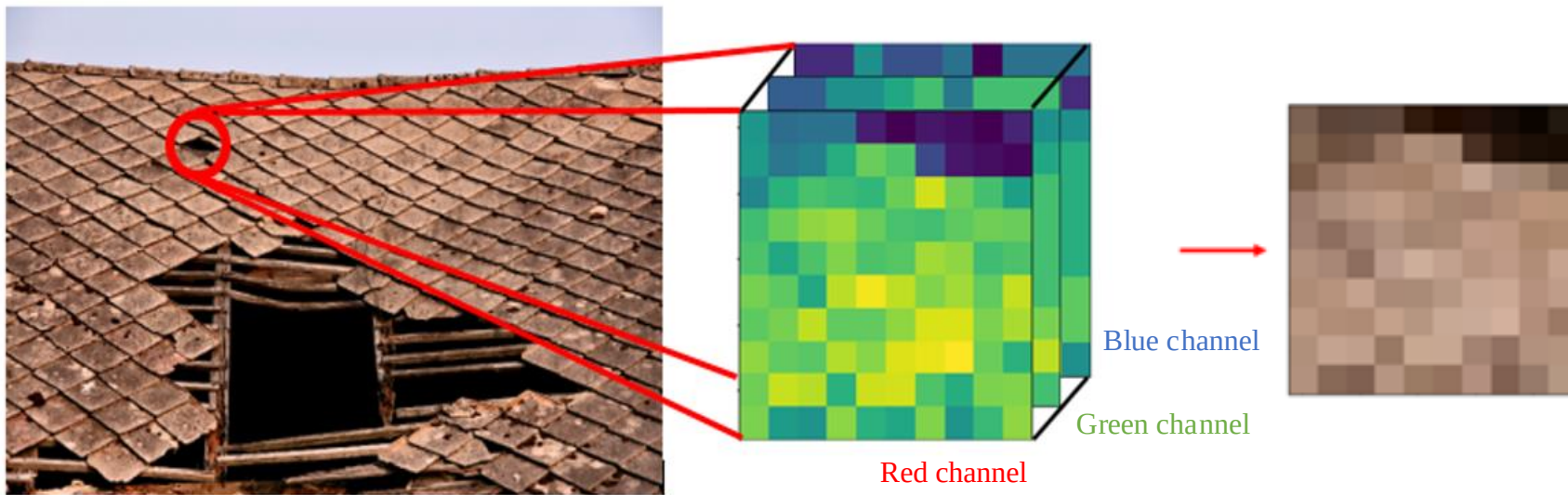
How computers interpret images?



Grayscale image

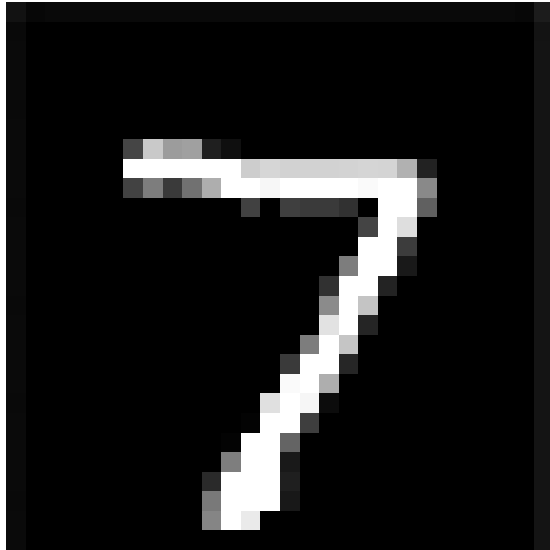
How computers interpret images?

When it comes to real-life applications, most images are in fact a 3D tensor with width, height, and 3 channels (R,G,B) as dimensions.



Colour image

Image as digits (pixel values)



- This image is from MNIST dataset. It has been compressed to pixel size 28x28 (784 pixels)
- Upload this image to <https://www.dcode.fr/digits-image> and get its corresponding pixel values.

IMAGE DIGITIZER

★ IMAGE TO CONVERT INTO DIGITS seven.png [X]

The image will be converted to grayscale

☒ ORIGINAL SIZE (NOT RECOMMENDED FOR LARGE IMAGES)

☐ REDUCED SIZE (RECOMMENDED)

☐ CUSTOM WIDTH

★ NUMBER OF DIGITS/LEVEL OF GRAY AVAILABLE

★ ORDER OF DIGITS ☒ 0=BLACK, N=WHITE

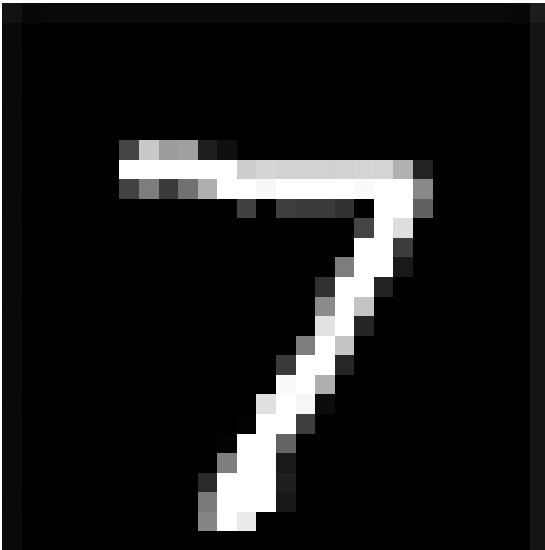
☐ 0=WHITE, N=BLACK

▶ CONVERT

See also: [Binary Image 0 1](#)

IMAGE FROM DIGITS CREATOR

Image as digits (pixel values)

[illegible]

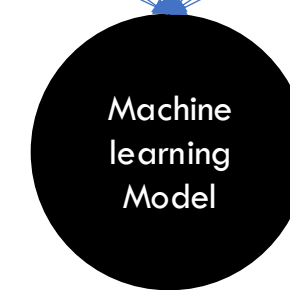
Neural network

This process of going from 2D to 1D vector is known as **flattening**. Because these networks cannot take 2D data as input.

X1	X2	X3	X4	X5
X6	X7	X8	X9	X10
X11	X12	X13	X14	X15
X16	X17	X18	X19	X20
X21	X22	X23	X24	X25

Input data

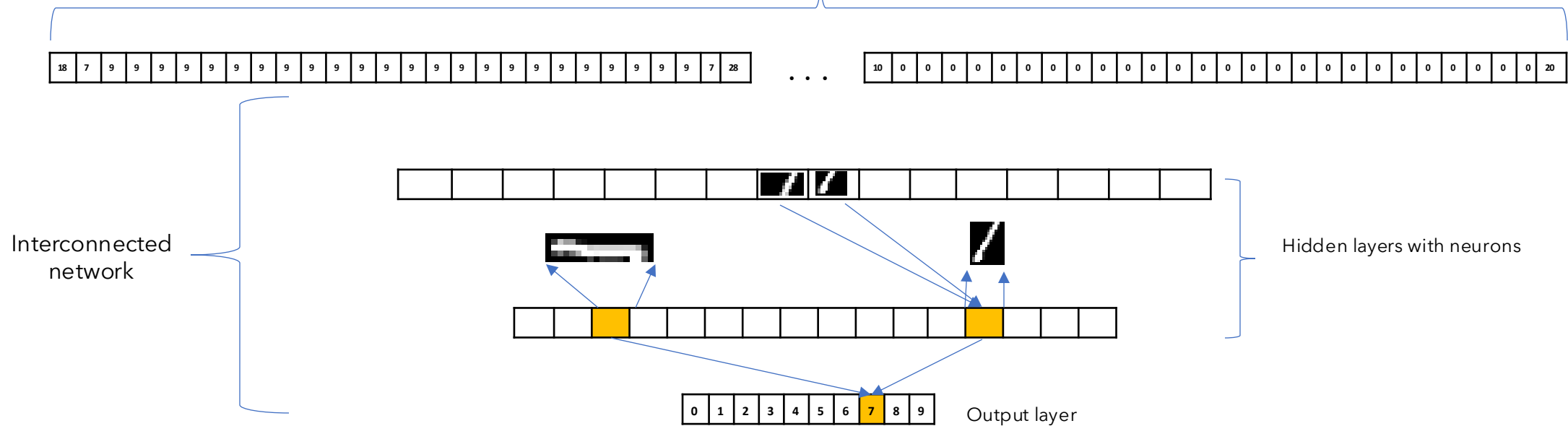
X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	...	X21	X22	X23	X24	X25
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Roof top Damaged
Yes/No

Input layer of the neural network

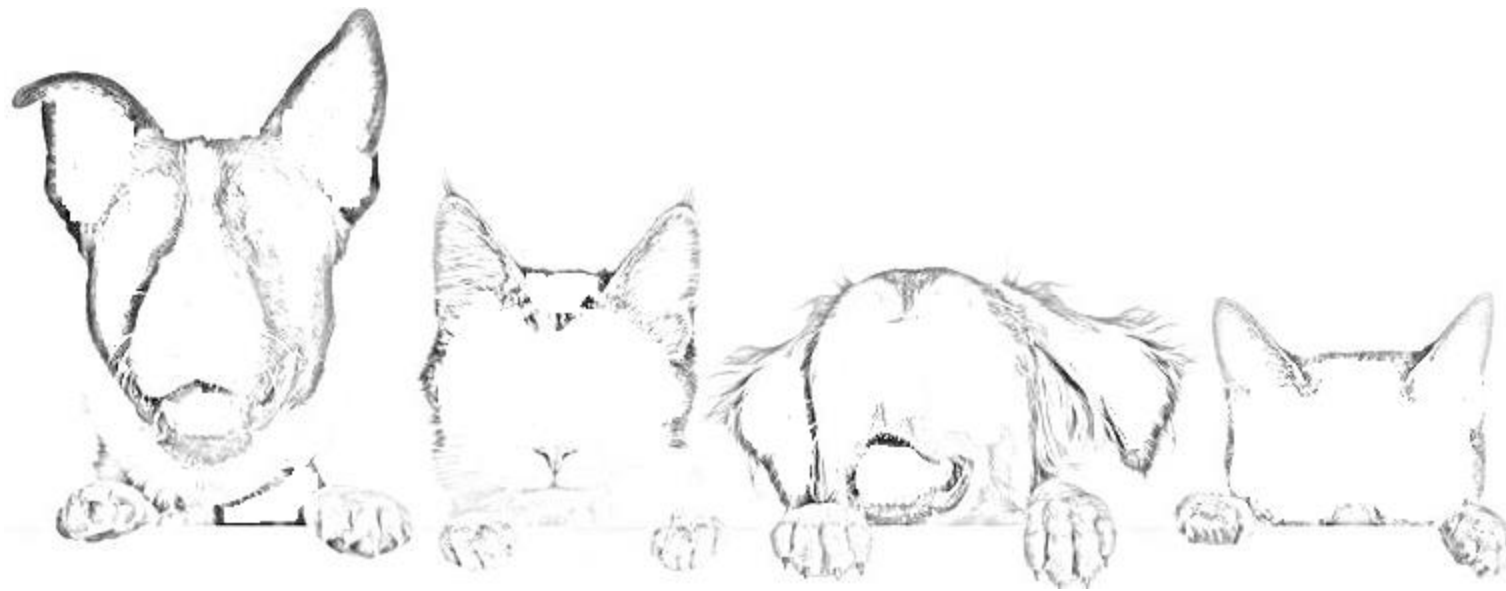
Total 784 (28x28)

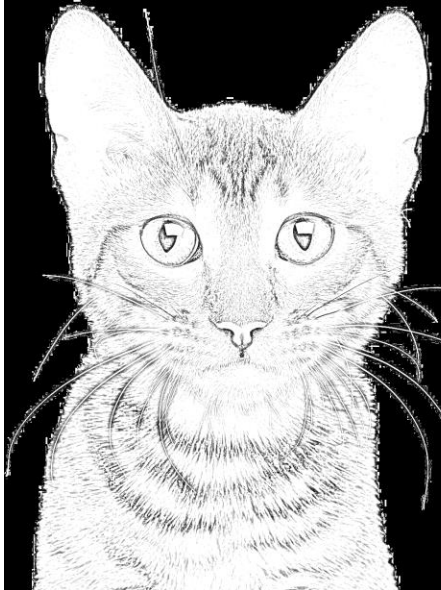


This process of going from 2D to 1D vector is known as **flattening**. Because these networks cannot take 2D data as input.



Initial Layers: Early layers will focus on basic features like lines and curves.





Intermediate Layers: As the network goes deeper, it learns more specific features, such as the presence of stripes (more prominent in tigers) and the overall body shape.

